

APPOINTMENTS	Yale University, Department of Statistics & Data Science, Assistant Professor Stanford University, Department of Mathematics, Science Fellow Harvard University, Department of Mathematics, NSF MPS Ascend Postdoctoral Fellow	July 2026–Present July 2023–June 2026 July 2022–June 2023
EDUCATION	Ph.D. in Mathematics, University of California, Berkeley Thesis: <i>Random Walks and Delocalization through Graph Eigenvector Structure</i> Advisors: Nikhil Srivastava and Luca Trevisan A.B. in Mathematics, Harvard University Secondary Field: The Classics Thesis: <i>Real Stable Polynomials: Description and Application</i> Thesis Advisor: Péter Csikvári	Aug. 2017–May 2022 Aug. 2011–May 2015
RESEARCH INTERESTS	Spectral theory of random and deterministic graphs: random matrix universality, eigenvector delocalization and quantum ergodicity, with connections to statistical physics, probability, and theoretical computer science.	
SELECTED HONORS AND FELLOWSHIPS	Semi-Plenary Speaker, Foundations of Computational Mathematics (FoCM 2026) Research featured in <i>Quanta Magazine</i>: “New Proof Settles Decades-Old Bet About Connected Networks” Stanford Science Fellowship NSF Math and Physical Sciences ASCEND Postdoctoral Fellowship NSF Graduate Research Fellowship Ford Foundation Predoctoral Fellowship UC Berkeley Math and Physical Sciences Mentorship Award Berkeley Chancellor’s Fellowship	2026 2025 2023 2022 2019 2019 2018 2017
SCIENTIFIC ORGANIZATION	Co-organizer, <i>Beyond Graph Laplacians</i>, Simons Institute for the Theory of Computing, UC Berkeley Organizer, Stanford Probability Seminar	Nov. 2026 (forthcoming) Winter 2024

PUBLICATIONS AND
PREPRINTS

15. An Alon–Boppana bound for the non-backtracking operator. Preprint, arXiv:2609.17529.
14. The Bethe–Hessian down to the percolation threshold. With Dingding Dong. arXiv:2608.16672, *Submitted*.
13. The Gaussian wave for graphs of finite cone type. With Amir Dembo. arXiv:2603.04386, *Submitted*.
12. Ramanujan property and edge universality of random regular graphs. With Jiaoyang Huang and Horng-Tzer Yau. arXiv:2412.20263, *Submitted*.
11. Arbitrary spectral edge of regular graphs. With Dingding Dong. *Advances in Mathematics* **499** (2026), 111047.
10. Optimal eigenvalue rigidity of random regular graphs. With Jiaoyang Huang and Horng-Tzer Yau. Permanent preprint, arXiv:2405.12161.
9. The spectral edge of constant degree Erdős–Rényi graphs. With Ella Hiesmayr. *Random Structures & Algorithms* **66**(3) (2025), e70011.
8. Nodal decompositions of a symmetric matrix. With John Urschel. *International Mathematics Research Notices* **2024**(7), 6224–6258.
7. Explicit two-sided unique-neighbor expanders. With Jun-Ting Hsieh, Sidhanth Mohanty, and Pedro Paredes. *Proceedings of the 56th Annual ACM SIGACT Symposium on Theory of Computing (STOC 2024)*.
6. Quantum ergodicity for periodic graphs. With Mostafa Sabri. *Communications in Mathematical Physics* **403** (2023), 1477–1509.
5. Many nodal domains in random regular graphs. With Shirshendu Ganguly, Sidhanth Mohanty, and Nikhil Srivastava. *Communications in Mathematical Physics* **401** (2023), 1291–1309.
4. The necessity of conditions for graph quantum ergodicity and Cartesian products with an infinite graph. *Comptes Rendus Mathématique* **360** (2022), 399–408.
3. Support of closed walks and second eigenvalue multiplicity of graphs. With Peter M. R. Rasmussen and Nikhil Srivastava. *Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing (STOC 2021)*, 396–407.
2. High-girth near-Ramanujan graphs with lossy vertex expansion. With Sidhanth Mohanty. *Proceedings of the 48th International Colloquium on Automata, Languages, and Programming (ICALP 2021)*.
1. A new algorithm for the robust semi-random independent set problem. With Hermish Mehta and Luca Trevisan. *Proceedings of the 31st Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2020)*, 738–746.

EXPOSITORY
WRITING

1. Predictive policing: a mathematical primer. With Joseph Johnson and Tian An Wong. *Notices of the American Mathematical Society* **71**(7) (2024).

INVITED CONFERENCE AND WORKSHOP LECTURES	Séminaire de Mathématiques Supérieures: <i>Growth, Asymptotics and Extremality in Geometry</i> , CRM/SLMath, Montréal, <i>Lecture Series</i>	June 2027 (forthcoming)
	Foundations of Computational Mathematics (FoCM 2026), <i>Semi-Plenary Speaker</i>	July 2026
	Oberwolfach Workshop on Random Matrices	June 2026
	University of Bristol Summer School, <i>Lecture Series</i>	June 2026
	Spectral Gaps Conference	Sep. 2025
	INFORMS Applied Probability Society Conference	June 2025
	Spectral Geometry in the Clouds	Mar. 2025, Mar. 2022
	IPAM Workshop on Free Entropy Theory and Random Matrices	Feb. 2025
	Harvard CMSA Conference on Graphical Models, Statistical Inference, and Algorithms (GRAMSIA)	May 2023
	IPAM Quantitative Linear Algebra Program	Dec. 2022, Dec. 2019
	Berkeley Edge Conference, <i>Keynote Speaker</i>	Nov. 2021
SELECTED INVITED SEMINAR TALKS AND COLLOQUIA	King's College London Mathematics Colloquium	Dec. 2026 (forthcoming)
	Columbia Probability Seminar	Sep. 2026
	Lehigh–Minnesota Probability Seminar	May 2026
	Caltech Combinatorics Seminar	Jan. 2026
	MIT Combinatorics Seminar	Apr. 2025, Feb. 2023
	Michigan Mathematics Colloquium	Dec. 2024
	Carnegie Mellon Probability/Math Finance Seminar	Sep. 2024
	Duke Probability Seminar	Sep. 2024
	University of Strasbourg Mathematical Physics Seminar	June 2024, Apr. 2022
	Texas A&M Mathematical Physics Seminar	Apr. 2024, Apr. 2022
	NYU Abu Dhabi Stochastics Seminar	Apr. 2024
	NYU Probability and Mathematical Physics Seminar	Sep. 2023
	IAS Computer Science and Discrete Mathematics Seminar	May 2023
	Georgia Tech Stochastics Seminar	Oct. 2021
	Carnegie Mellon Theory Seminar	Oct. 2021
	University of Washington CS Theory Seminar and Probability Seminar	Oct. 2021
SELECTED CONTRIBUTED TALKS	World Congress in Probability and Statistics	Aug. 2024
	International Congress of Mathematical Physics	July 2024
	International Colloquium on Automata, Languages, and Programming (ICALP)	July 2021
	Symposium on Theory of Computing (STOC)	June 2021
	Conference of Ford Fellows	Oct. 2020, Oct. 2019
	Symposium on Discrete Algorithms (SODA)	Jan. 2020

TEACHING EXPERIENCE	Principal Instructor, Math 136: <i>Stochastic Processes</i> , Stanford University	Fall 2025
	Principal Instructor, Math 107: <i>Graph Theory</i> , Stanford University	Winter 2024
	Lecturer, Boston University Prison Education Program, MCI-Framingham Prison	2022–2023
	Instructor (Lead Instructor, 2018–2022), Precalculus and Intermediate Algebra, Mount Tamalpais College (formerly the Prison University Project), San Quentin State Prison	2018–2023
MENTORING AND SERVICE	Mentor, Harvard Math Includes — mentored five undergraduates from underrepresented backgrounds	2022–2023
	Member, UC Berkeley Mathematics Equity and Inclusion Committee	2021–2022
	Ambassador, UC Berkeley STEM First Year Initiative — mentored three first-year minority STEM Ph.D. students	2019–2022
	Mentor, Berkeley Getting into Graduate School — advised two students on program selection and application materials	2019–2020
PROFESSIONAL EXPERIENCE	Research Intern, Microsoft Research, Machine Learning Foundations Group, <i>Redmond, WA</i>	May 2022–Aug. 2022
	Trading Analyst, Goldman Sachs & Co., <i>New York, NY</i>	July 2015–May 2017